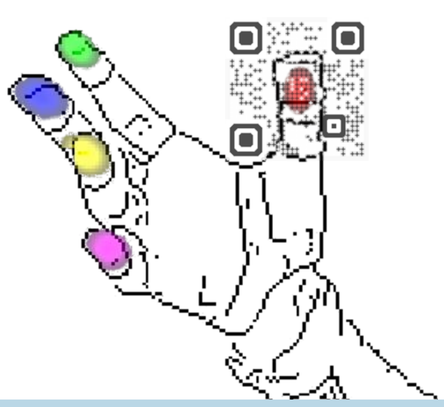


Rule of Thumb: Deep derotation for improved fingertip detection

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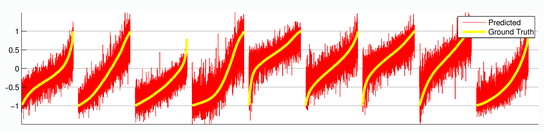
Problem statement

We propose DeROT, a method for in-plane derotation of depth images using a deep Convolutional Neural Network (CNN). The method is aimed at normalizing out the global effects of rotation on highly articulated motion of deforming geometric surfaces such as hands. The proposed method reduces the complexity of learning in the space of articulated poses. We demonstrate this by using two different state-of-the-art learning based hand pose estimation methods (RDT [1] and CNN [2]) applied to per-frame fingertip detection. To support our approach we describe a new pipeline for quickly building a very large training database using high accuracy magnetic sensor annotation (TrakStar [3]) and labeling of objects imaged by a depth camera.

DeROT: Deep derotation

In order to derotate an image to a canonical form we need to produce an in-plane derotation angle. We do this in a two-step process.

First, a Convolutional Neural Network (adapted from [2]) is trained to predict the 9 coefficients of a rotation matrix which we then project back to $SO(3)$.

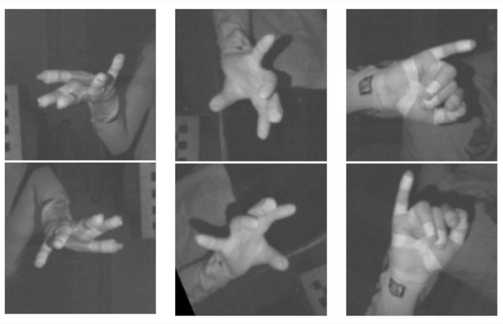


Next, we apply an insight which we call "Rule of Thumb" which is based on the observation that there is almost always an in-plane rotation which can be applied to an image of the hand that forces the thumb to be on the right side of the image.

Algorithm 1 Derotation procedure

procedure DEROTATE(R)
  $r_{align} \leftarrow \text{argmax}_{r_i \in \{r_1, r_2, r_3\}} \| (0,0,1) \cdot r_i \|$
  if $r_{align} = r_2$ (thumb aligned axis) **then**
    $\alpha \leftarrow atan2(r_{3x}, r_{3y}) + 90 + \begin{cases} 180 & \text{if } r_{2z} \le 0 \\ 0 & \text{if } r_{2z} > 0 \end{cases}$
  else
    $\alpha \leftarrow atan2(r_{2x}, r_{2y}) + 90$
  end if
  Return α
end procedure

We call this two-step procedure DeROT. By derotating both our training images and new test images we are able to partially disentangle the effect of global hand rotation in the input space.



Contributions

- DeROT: A novel image derotation procedure for partial global orientation normalization.
- Useful modifications to RDT and CNN based per-frame fingertip detection.
- HandNet: The largest, fully annotated, user-facing database of depth images of hands captured by a RealSense RGBD sensor.

HandNet: A new hand dataset

www.cs.technion.ac.il/~twerd/HandNet

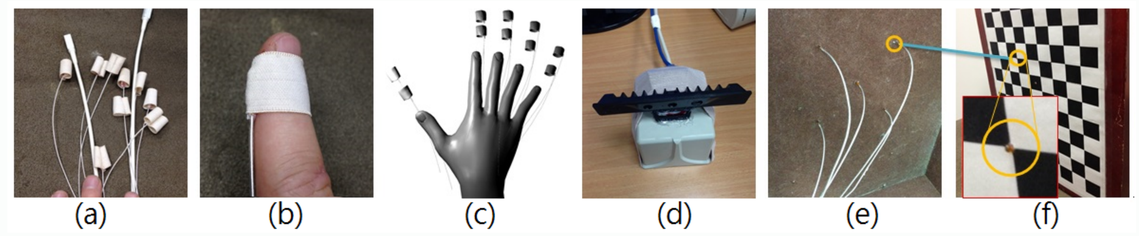


Figure 1: a) 2mm magnetic sensors. b) A fingertip sensor inside the inner seam. c) Virtual model used for planning a multi-sensor setup. We only use 5 sensors. d) A RealSense camera rigidly fixed to the TrakStar transmitter. e) The back of the wooden calibration board where the glass sensor housings are firmly pushed through. f) The front of the calibration board where the glass sensor housings are visible on the corners as seen in the inset.

Using a Realsense RGBD camera and high speed, high accuracy TrakStar magnetic sensors we recorded a new dataset of hand images with:

- 10 participants (6 male, 4 female)
- 202928 total images for training, 10000 for testing, 2773 for validation/tuning.
- Full $6D$ global hand pose available as well as individual $6D$ pose available for every fingertip. Per pixel fingertip labels also available.

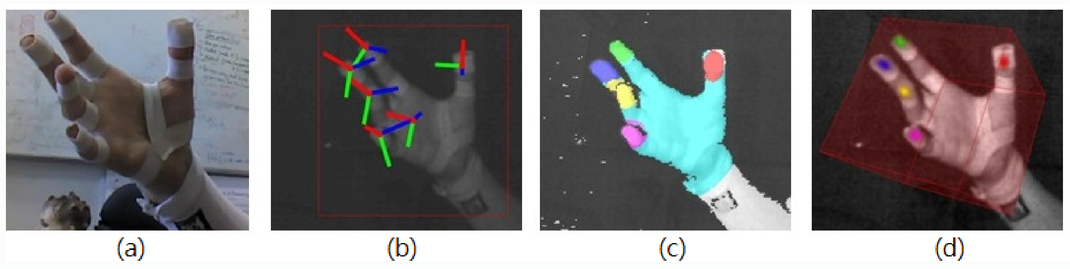
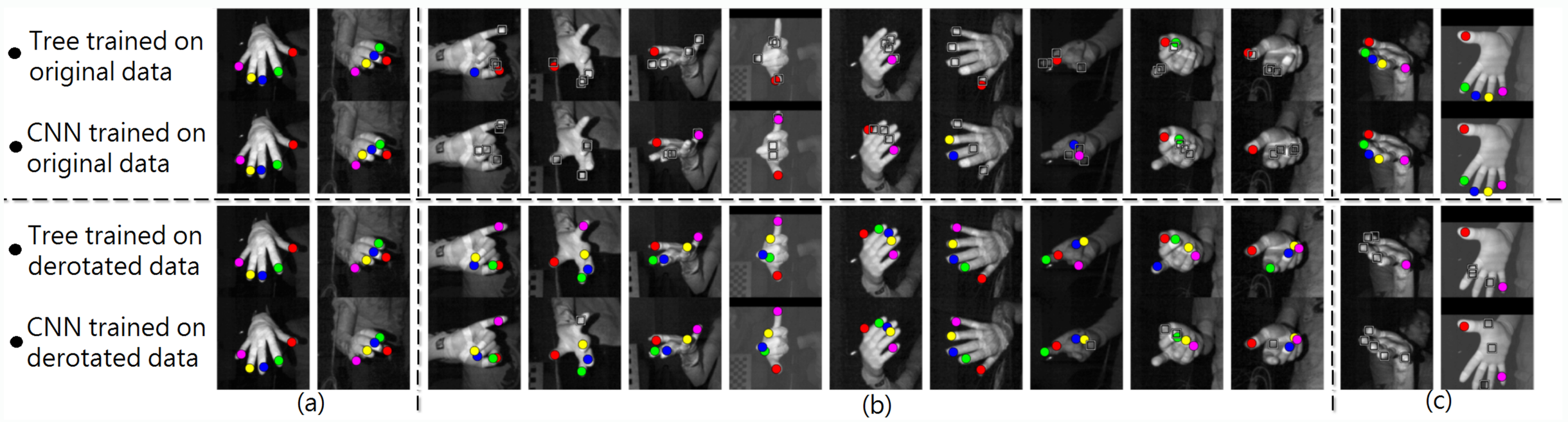


Figure 2: The available data annotations after calibration. a) Color image (not used). b) The RGB axes indicate the measured location and orientation of each fingertip and the back of the palm. c) IR image (not used) overlaid with the raycasted labels. d) IR image overlaid with the generated heatmaps per fingertip and the global orientation of the hand represented as an oriented bounding box (not used).

Visualizing results on HandNet



References

[1] C. Keskin, F. Kirac, Y. Emre Kara, and L. Akarun Real time hand pose estimation using depth sensors, In *Consumer Depth Cameras for CV, 2013*

[2] J. Tompson, M. Stein, Y. Lecun, and K. Perlin Real-time continuous pose recovery of human hands using convolutional networks., In *ACM TOC, 33, 2014*

[3] Ascension TrakStar. `http://www.ascension-tech.com/`, 2013

Acknowledgements

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Results

Test set derotation method	None mP | mAP	(a) Procrustes mP | mAP	(b) DeROT mP | mAP	(c) Oracle mP | mAP
HandNet				
RDT trained on non-derotated data	0.51 | 0.79	0.49 | 0.77	**0.55** | **0.85**	**0.60** | **0.87**
RDT trained on derotated data	0.32 | 0.60		**0.63** | **0.88**	**0.75** | **0.95**
CNN trained on non-derotated data	0.44 | 0.73	0.42 | 0.73	**0.46** | **0.77**	**0.50** | **0.79**
CNN trained on derotated data	0.30 | 0.59		**0.61** | **0.88**	**0.74** | **0.95**
NYUHands				
RDT trained on non-derotated data	0.51 | 0.75	0.47 | 0.73	**0.58** | **0.84**	**0.61** | **0.86**
RDT trained on derotated data	0.35 | 0.58		**0.63** | **0.88**	**0.68** | **0.89**
CNN trained on non-derotated data	0.38 | 0.70	0.36 | 0.69	**0.46** | **0.80**	**0.48** | **0.81**
CNN trained on derotated data	0.23 | 0.42		**0.49** | **0.72**	**0.53** | **0.73**